

Answer, December 19, 2016.

Course name: Network Optimization

Course no: 42115

Shortest path problem

Answer 1 Since we have negative edge weights, we cannot use Dijkstra. Instead we can use Bellman-Ford. However, since there are no cycles in the graph, the fastest method is to order the nodes in topological order, and use dynamic programming. ■

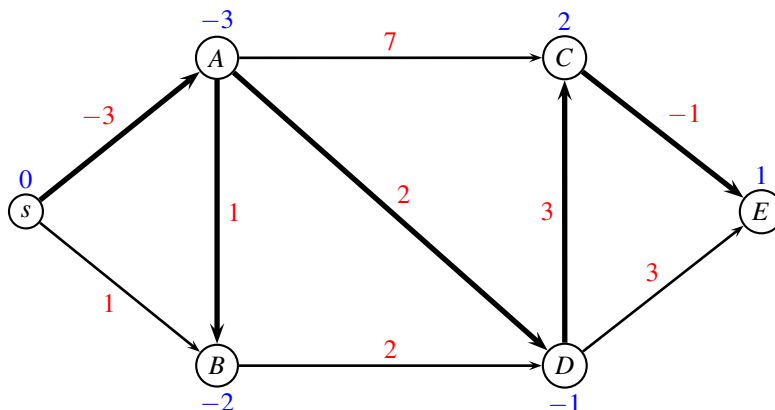
Answer 2 We start by sorting the nodes in topological order A, B, D, C, E . We now run dynamic programming using recursion

$$d(j) = \min_{(i,j) \in E} d(i) + w_{ij}$$

where w_{ij} are the edge weights. This gives

$$\begin{aligned} d(s) &= 0 \\ d(A) &= \min\{0 - 3\} = -3 \\ d(B) &= \min\{0 + 1, -3 + 1\} = -2 \\ d(D) &= \min\{-3 + 2, -2 + 2\} = -1 \\ d(C) &= \min\{-3 + 7, -1 + 3\} = 2 \\ d(E) &= \min\{2 - 1, -1 + 3\} = 1 \end{aligned}$$

The solution becomes:



The bold edges mark the shortest path tree.

Alternative solution

Running Bellman-Ford takes longer time, but is a correct approach. We have the potentials:

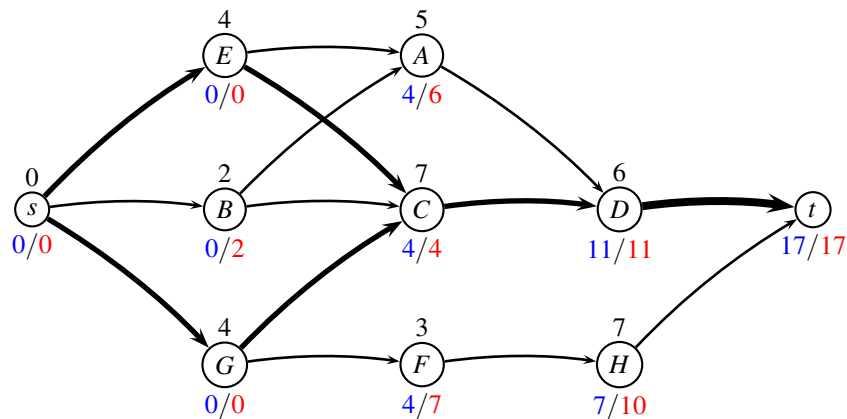
iteration	s	A	B	C	D	E
0	0	∞	∞	∞	∞	∞
1	0	-3	-2	2	-1	2
2	0	-3	-2	2	-1	1
3	0	-3	-2	2	-1	1

In principle we should run $n - 1 = 6 - 1$ iterations, but since the last two iterations do not change the potentials, we can stop. ■

Critical path problem

Answer 3 a possible topological ordering is E, B, G, A, C, F, D, H ■

Answer 4 We insert a new node s and connect it to the three activities E, B, G with no predecessors. We insert a new node t and connect it to the two activities D, H with no successors. This gives the following graph (duration of activities is indicated with black on the nodes):



Now we solve the problem

- *Forward procedure:* Starting at time zero, calculate the earliest time U_j each job j can be started $U_j = \max_{(i,j) \in E} \{U_i + p_i\}$. (marked with blue, above)
- *Makespan:* $T = U_t$.
- *Backward procedure:* Starting at time equal to the makespan, calculate the latest time V_j each job j can be started so that this makespan is realized. $V_i = \min_{(i,j) \in E} \{V_j - p_i\}$ (marked with red, above)

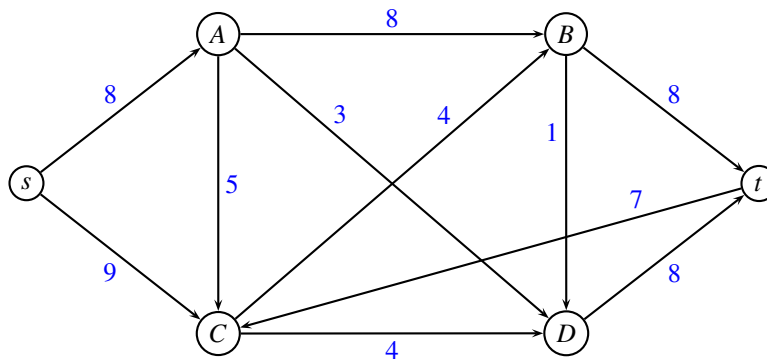
The critical path schedule solution for this project is given below:

Activity	Duration	Earliest start time U_j	Latest start time V_j
s	0	0	0
A	5	4	6
B	2	0	2
C	7	4	4
D	6	11	11
E	4	0	0
F	3	4	7
G	4	0	0
H	7	7	10
t	0	17	17

The makespan is $T = 17$.

The critical path is found for activities where earliest start time U_j is equal to the latest start time V_j in the above table. This means projects C, D, E, G . The critical path(s) are marked with bold edges in the above graph. ■

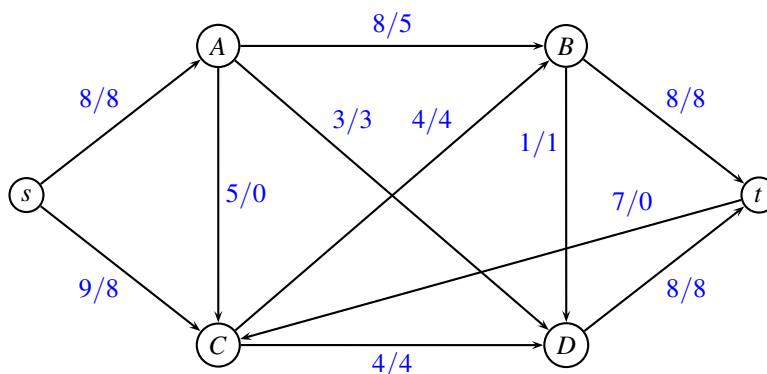
Maximum flow



Answer 5 We use the paths:

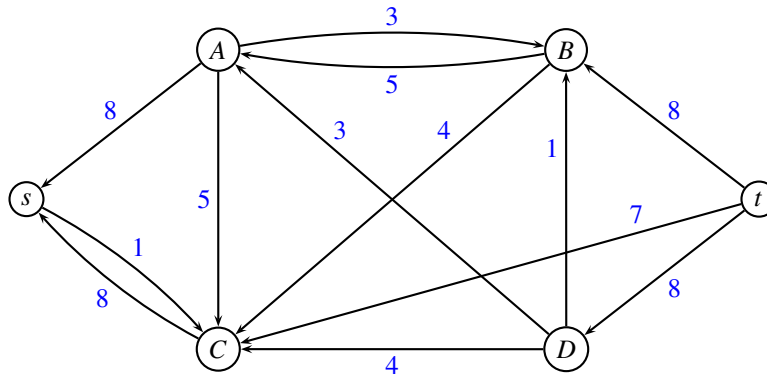
path	capacity
sABt	8
sCDt	4
sCBADt	3
sCBDt	1

This results in the following solution (edge labels are: capacity/flow):



The maximum flow is 16. ■

Answer 6 The residual graph is



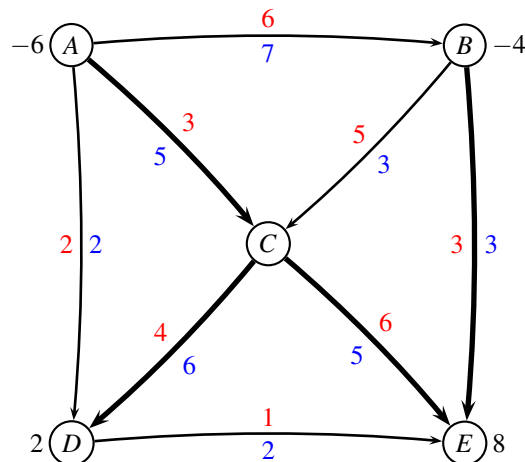
A minimum cut is found by marking all nodes that are reachable from s in the residual graph. This gives the sets $\{s, C\}$ and $\{A, B, D, t\}$. The capacity of the cut is $8 + 1 + 3 + 4 = 16$.

Alternative solution

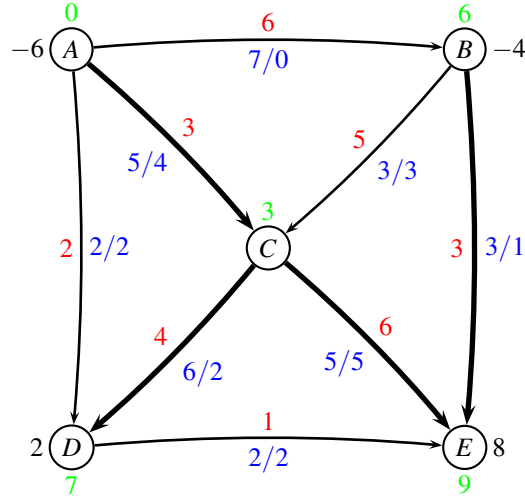
Another minimum cut (found by inspection) is $\{s, A, B, C\}$ and $\{D, t\}$. ■

Minimum cost flow problem

Answer 7 We have the following network



We are told that $L = \{(A, B)\}$ and $U = \{(A, D), (B, C), (D, E)\}$. The flow on edges L is at the lower bound, while the flow on edges in U is at the upper bound. Based on this information, we can find the flow as



This is our initial flow. The cost of the flow is

$$z = 3 \cdot 4 + 2 \cdot 2 + 5 \cdot 3 + 3 \cdot 1 + 4 \cdot 2 + 6 \cdot 5 + 1 \cdot 2 = 74$$

■

Answer 8 Calculating the distances from A to all nodes, only using the edges in T gives the dual variables u_i , indicated with green on the figure.

We can now calculate reduced costs of all edges not in T using the equation $\bar{c}_{ij} = c_{ij} + u_i - u_j$. This gives

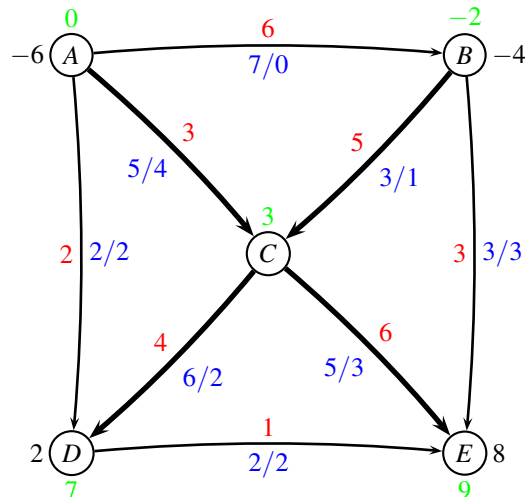
$$\begin{aligned}\bar{c}_{AB} &= 6 + 0 - 6 = 0 \quad (L) \\ \bar{c}_{AD} &= 2 + 0 - 7 = -5 \quad (U) \\ \bar{c}_{BC} &= 5 + 6 - 3 = 8 \quad (U) \\ \bar{c}_{DE} &= 1 + 7 - 9 = -1 \quad (U)\end{aligned}$$

For edges $(i, j) \in L$ the reduced cost $\bar{c}_{ij}$ should be negative. For edges $(i, j) \in U$ the reduced cost $\bar{c}_{ij}$ should be positive.

Therefore, we pivot on edge (B, C) with cycle C, B, E, C . The maximum number of units we can send is

$$\theta = \min\{3, 2, 5\} = 2$$

Sending two units along the cycle gives the graph



Calculating the distances from A to all nodes, only using the edges in T gives the dual variables u_i , indicated with green on the figure.

We can now calculate reduced costs of all edges not in T

$$\begin{aligned}\bar{c}_{AB} &= 6 + 0 - (-2) = 8 & (L) \\ \bar{c}_{AD} &= 2 + 0 - 7 = -5 & (U) \\ \bar{c}_{BE} &= 3 + (-2) - 9 = -8 & (U) \\ \bar{c}_{DE} &= 1 + 7 - 9 = -1 & (U)\end{aligned}$$

For edges $(i, j) \in L$ the reduced cost $\bar{c}_{ij}$ should be negative. For edges $(i, j) \in U$ the reduced cost $\bar{c}_{ij}$ should be positive. This is not the case, so we stop.

The cost of the optimal solution is:

$$z = 3 \cdot 4 + 2 \cdot 2 + 5 \cdot 1 + 3 \cdot 3 + 4 \cdot 2 + 6 \cdot 3 + 1 \cdot 2 = 58$$

Since we started from a solution with $z = 74$ and sent 2 units along a cycle with reduced cost -8 the optimal solution corresponds to what we would expect. ■

Column generation

Answer 9 We have the dual solution given by $y_{AB} = -10$, while all other $y_{ij} = 0$. Moreover, we have $\bar{y}_1 = 48, \bar{y}_2 = 21, \bar{y}_3 = 36$.

We calculate the reduced costs using the equation:

$$\bar{c}_r^k = \sum_{(i,j) \in r} d_k(c_{ij} - y_{ij}) - \bar{y}_k$$

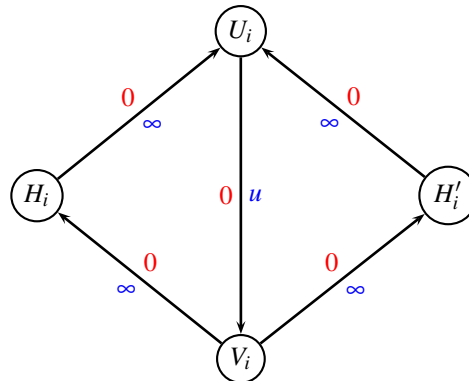
This gives the following table:

number	commodity k	route r	d_k	$\sum_{(i,j) \in r} c_{ij}$	$\sum_{(i,j) \in r} (c_{ij} - y_{ij})$	$d_k \sum_{(i,j) \in r} (c_{ij} - y_{ij})$	$\bar{y}_i$	$\bar{c}_r^k$
1	1	ABC	3	6	16	48	48	0
2	1	ABDC	3	20	30	90	48	42
3	1	ADBC	3	14	14	42	48	-6
4	1	ACD	3	16	16	48	48	0
5	2	EABC	1	11	21	21	21	0
6	2	EABDC	1	25	35	35	21	14
7	2	EADBC	1	19	19	19	21	-2
8	2	EADC	1	21	21	21	21	0
9	3	BCDEA	3	8	8	24	36	-12
10	3	BDEA	3	12	12	36	36	0

The most negative reduced cost is route $r = 9$ with reduced cost -12 . ■

Modeling a hub

Answer 10 We use the modeling trick from assignment 17.21 in Ahuja, Orlin, Magnanti *Network Flows*: The edges between H_i and H'_i are replaced by the construction



Costs are red, and capacities are blue. Here, u is the capacity of the crane (8 or 5). This is done for all pairs (H_1, H'_1) , (H_2, H'_2) , and (H_3, H'_3) . ■

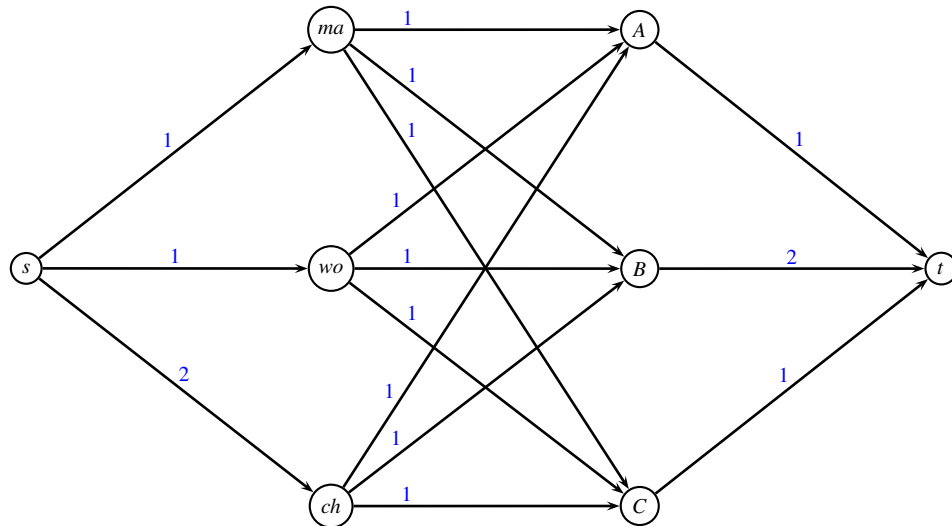
Census rounding

Answer 11 We are given a matrix of fractional values:

Table 3				
Country	A	B	C	Total
grown-up men	0,1	0,7	0,2	1,0
grown-up women	0,4	0,6	0,0	1,0
children	0,5	0,7	0,8	2,0
total	1,0	2,0	1,0	4,0

Since there is no objective function associated with the problem, we can formulate it as a maximum flow problem. Rows in the matrix (men, women, children) correspond to nodes ma , wo , ch , and columns in the matrix correspond to the nodes A , B , C . Edges with capacity 1 are added between each row and column node. Row nodes are connected to s with edges having a capacity corresponding to the demanded sum in the row. Column nodes are connected to s with edges having a capacity

corresponding to the demanded sum in the column.



Notice that an optimal solution always will have integral flow since the problem is TU.

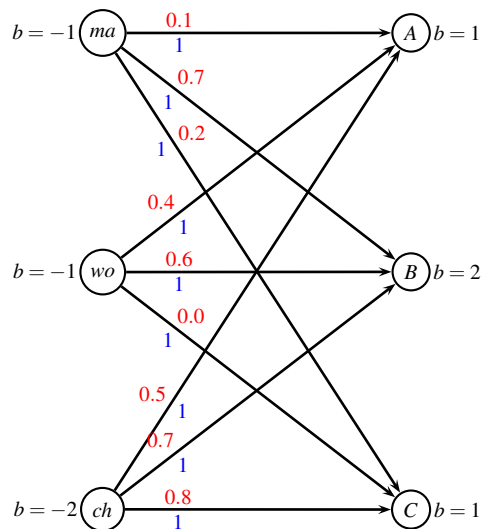
If there is a flow of 1 on an edge between a row node and a column node, then the corresponding entry in the matrix should be rounded up to 1. Otherwise it should be rounded down to 0.

Alternative solution

We can also model the problem as a minimum-cost-flow problem.

We construct a bi-partitate graph as follows: Rows in the matrix (men,women,children) correspond to nodes *ma*, *wo*, *ch*, and columns in the matrix correspond to the nodes *A*,*B*,*C*. Edges with capacity $u_{ij} = 1$ are added between each row node and column node.

Since there is no objective function associated with the problem, we can choose any edge costs c_{ij} but a value of $c_{ij} = 1 - f_{ij}$ where f_{ij} is the fractional value in Table 3, can be used.

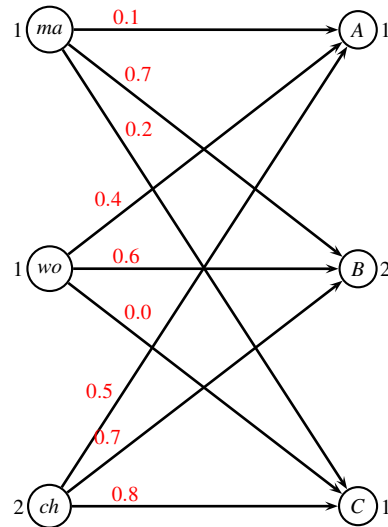


Notice that an optimal solution always will have integral flow since the problem is TU.

If there is a flow of 1 on an edge between a row node and a column node, then the corresponding entry in the matrix should be rounded up to 1. Otherwise it should be rounded down to 0.

(Wrong) alternative solution

One could try to model the problem as a transportation problem.



Since there is no objective function associated with the problem, we can again choose any edge costs c_{ij} . A value of $c_{ij} = 1 - f_{ij}$ where f_{ij} is the fractional value in Table 3, can be used.

Modeling the problem as a transportation problem will not work in general, since there are no capacities on the edges, and hence we can get a flow larger than one on edges. In the above example we could thus find a solution with a flow of 2 from *ch* to *B*. This means that the fractional value should be round up twice, which is not allowed. ■

Answer 12 In the general problem we are given the matrix

Table 1

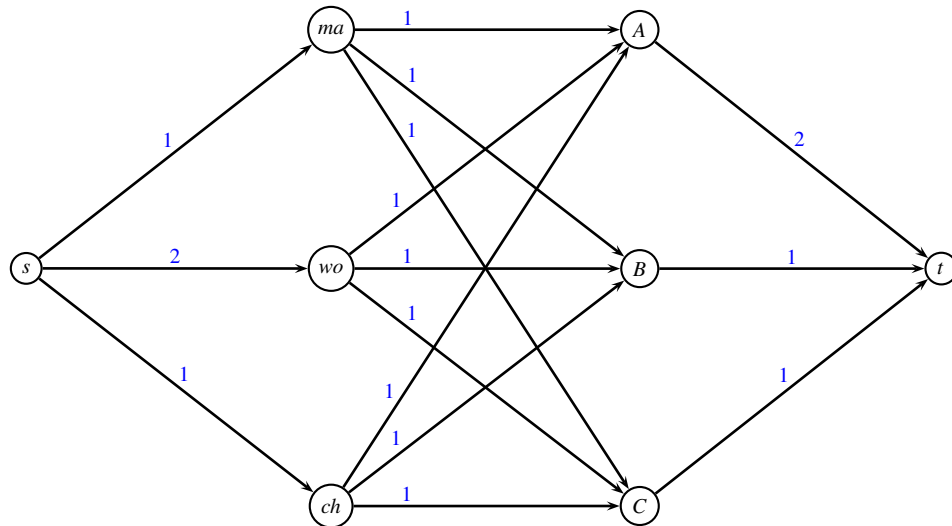
Country	A	B	C	Total
grown-up men	11,6	9,4	3,0	24,0
grown-up women	12,7	10,5	3,8	27,0
children	1,7	2,1	0,2	4,0
total	26,0	22,0	7,0	55,0

We first subtract the integer part from each entry, and subtract it from the row and column sums. This gives the matrix

Table 1*

Country	A	B	C	Total
grown-up men	0,6	0,4	0,0	1,0
grown-up women	0,7	0,5	0,8	2,0
children	0,7	0,1	0,2	1,0
total	2,0	1,0	1,0	4,0

This leads to the following maximum flow problem



Notice that an optimal solution always will have integral flow since the problem is TU.

If there is a flow of 1 on an edge between a row node and a column node, then the corresponding entry in the matrix should be rounded up, otherwise it should be rounded down.

Alternative solution

We can also model the problem as a minimum-cost-flow problem. The same principles as in previous answer are used. ■